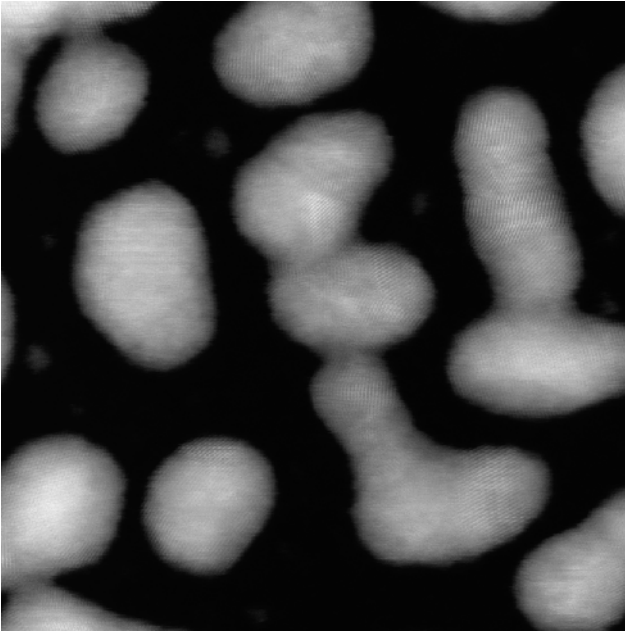


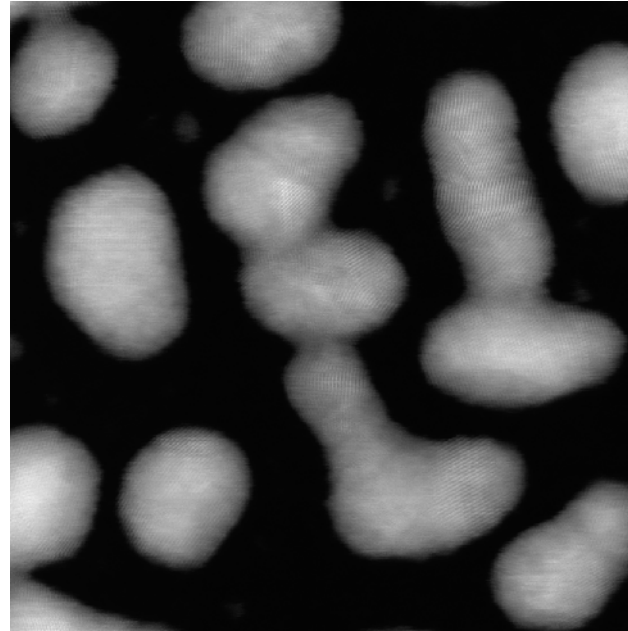
EDS Au (Gold) — Drift Correction Report

File 0041 vs Reference 0042+0043

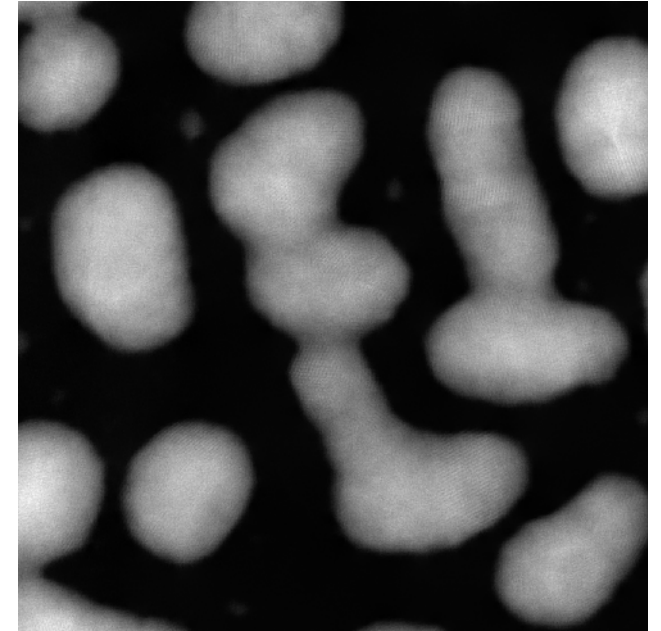
HAADF 0041 — raw



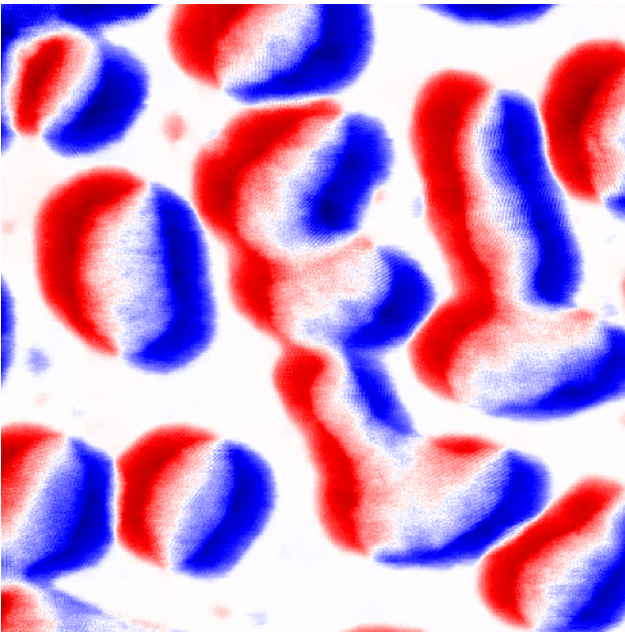
HAADF 0041 — after affine



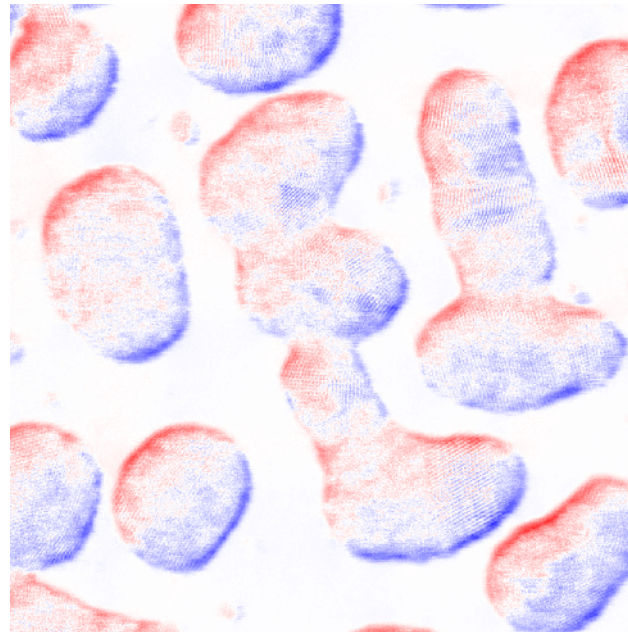
HAADF ref (0042+0043)



Difference: ref — raw



Difference: ref — after affine



Dataset: EDS Au (Gold)
File 0041: HAADF 1024×1024
EDS cube: (1024, 1024, 4096)
Reference: HAADF 0042+0043
(drift-corrected, downsampled)

Affine drift: +0.020 / +0.016 px/line
Total: ~26 px over 1024 rows

Baseline (raw):

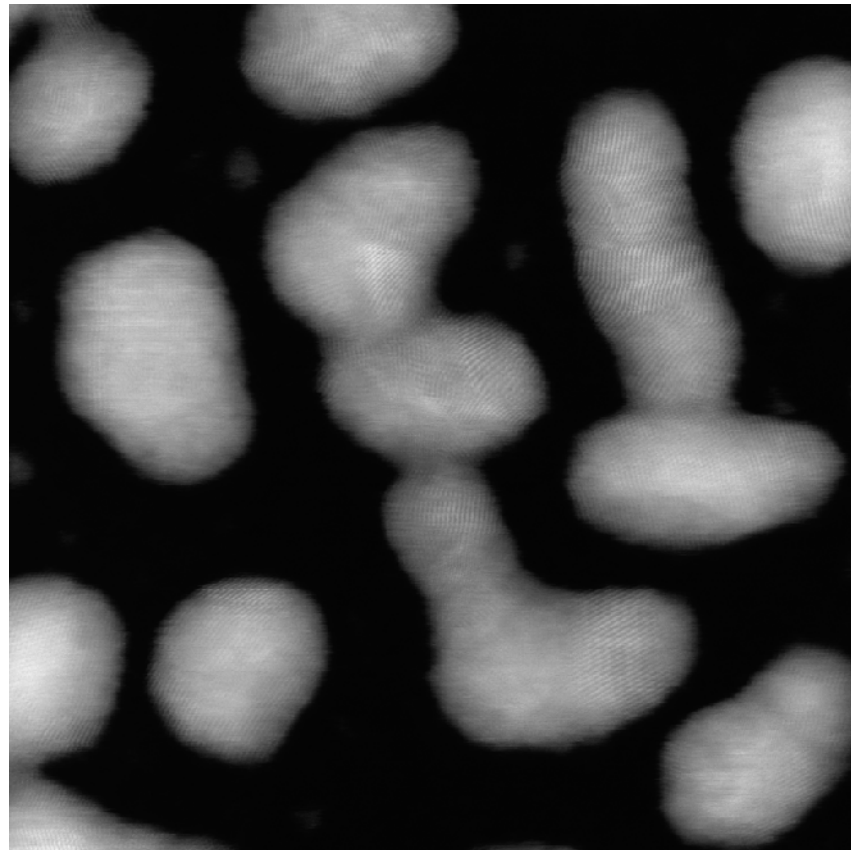
NCC 0.4080
MAE 0.7757
RMSE 1.0881

After affine:

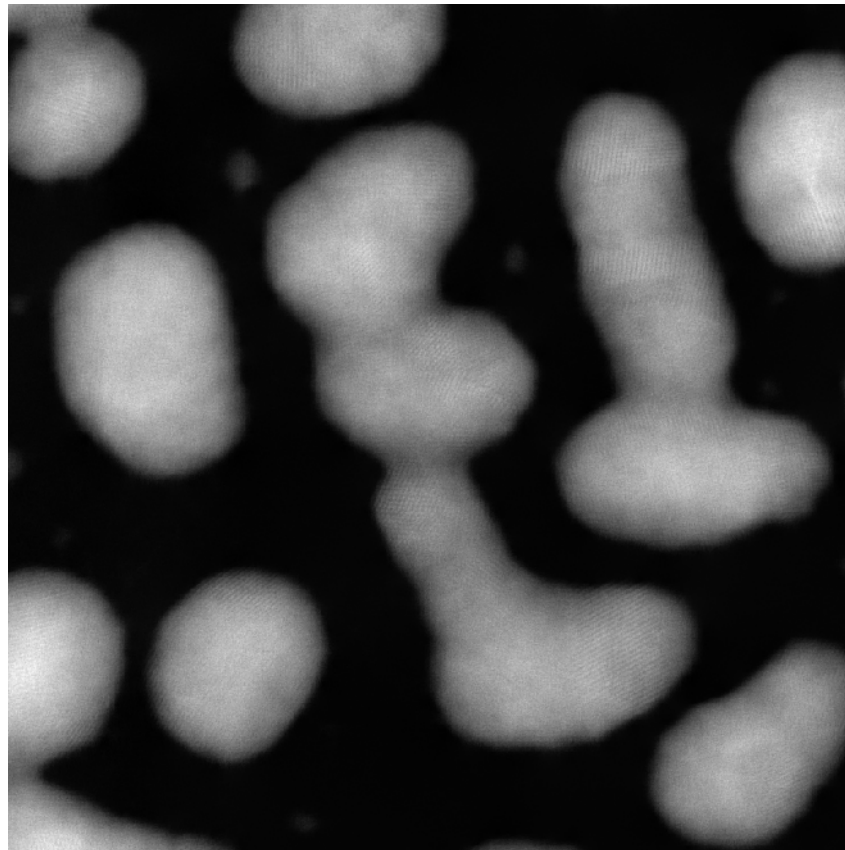
NCC 0.9718
MAE 0.1477
RMSE 0.2373

LBFGS $\sigma=8$ step=0.8 iters=12 (original baseline)
NCC 0.9822 MAE 0.1122 RMSE 0.1887 | nonrigid max shift 57.0 px | 0.9s

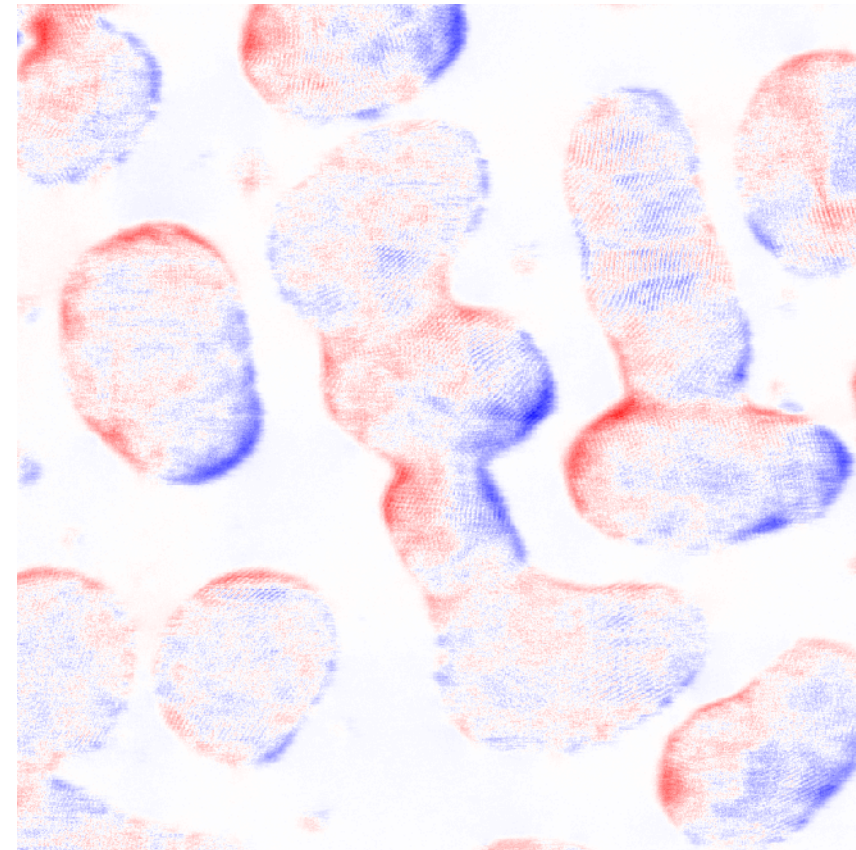
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

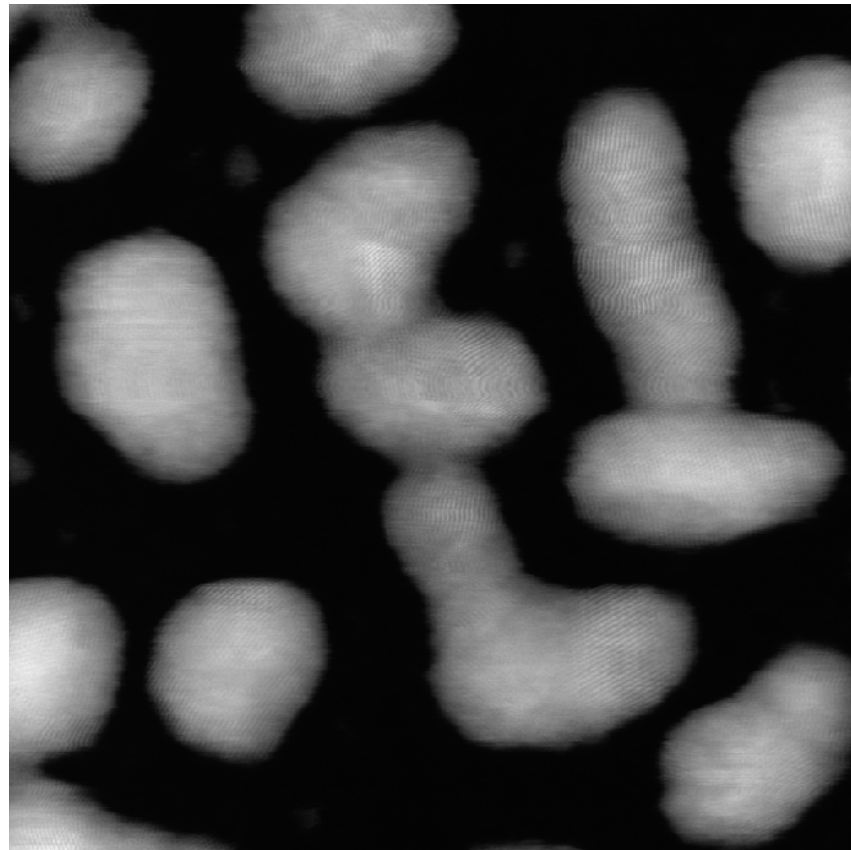


Difference: ref — corrected

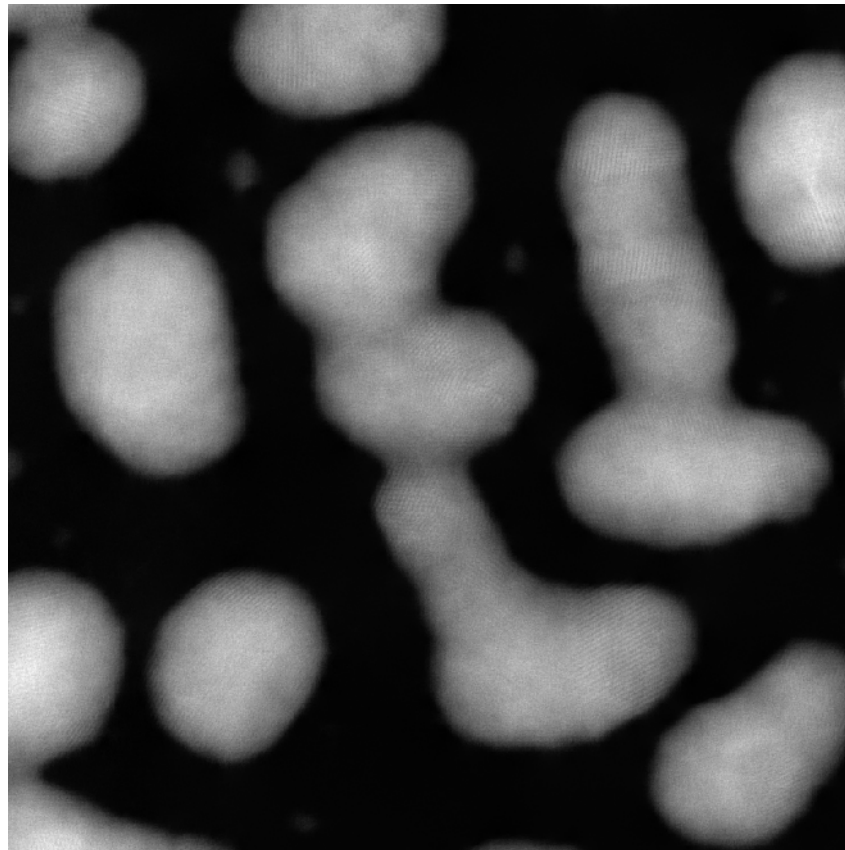


LBFGS $\sigma=2$ step=0.8 iters=12
NCC 0.9840 MAE 0.1066 RMSE 0.1791 | nonrigid max shift 58.6 px | 0.4s

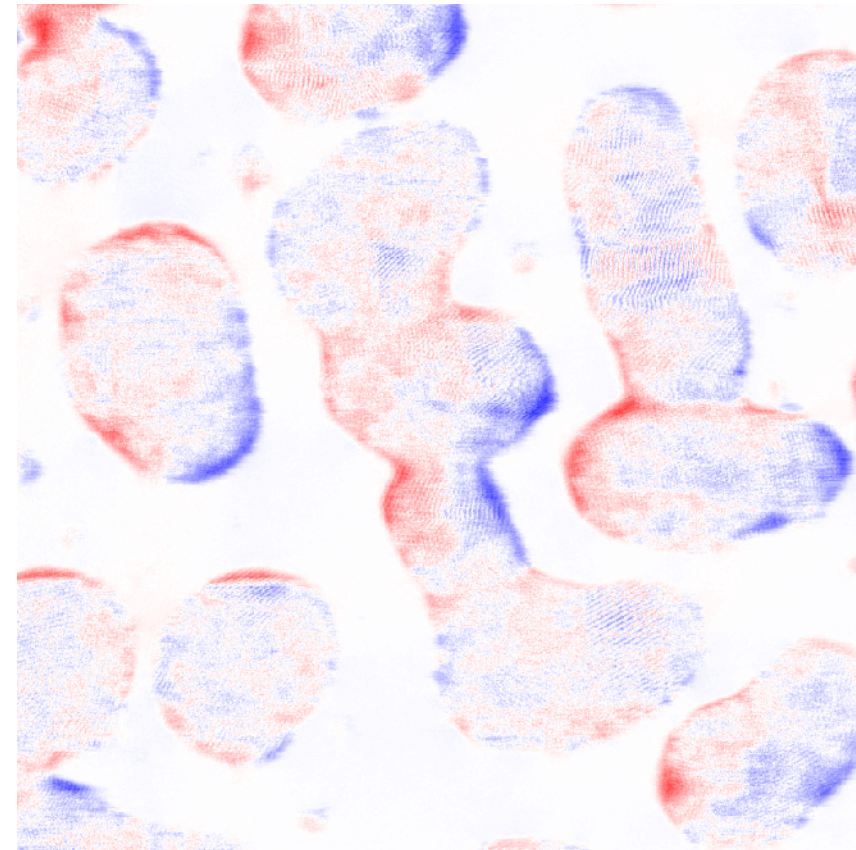
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

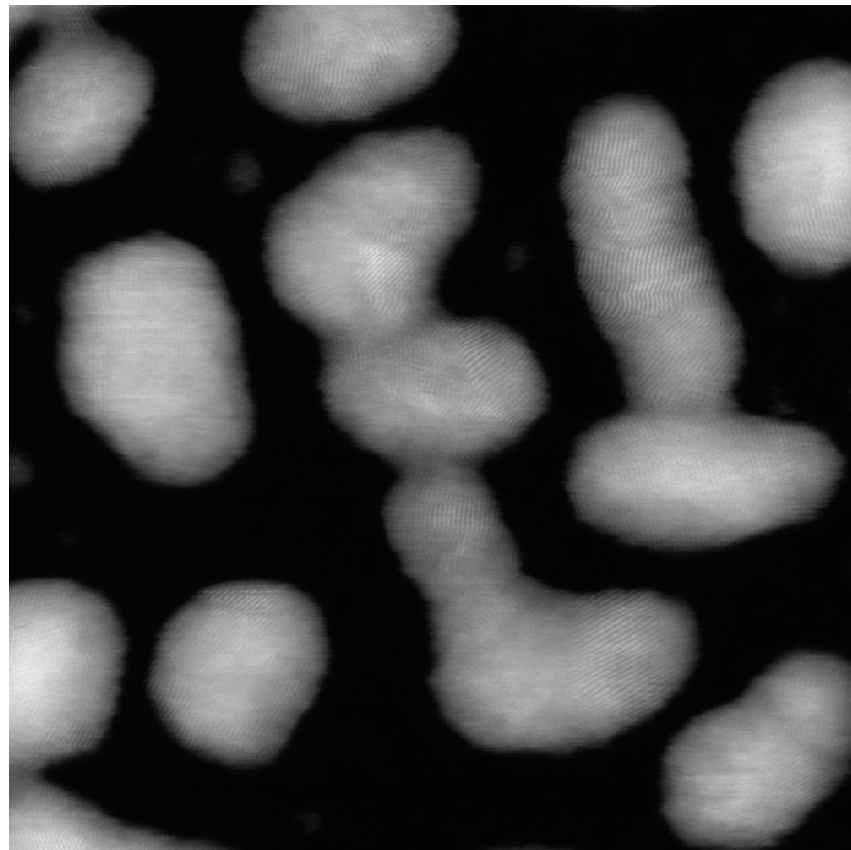


Difference: ref — corrected

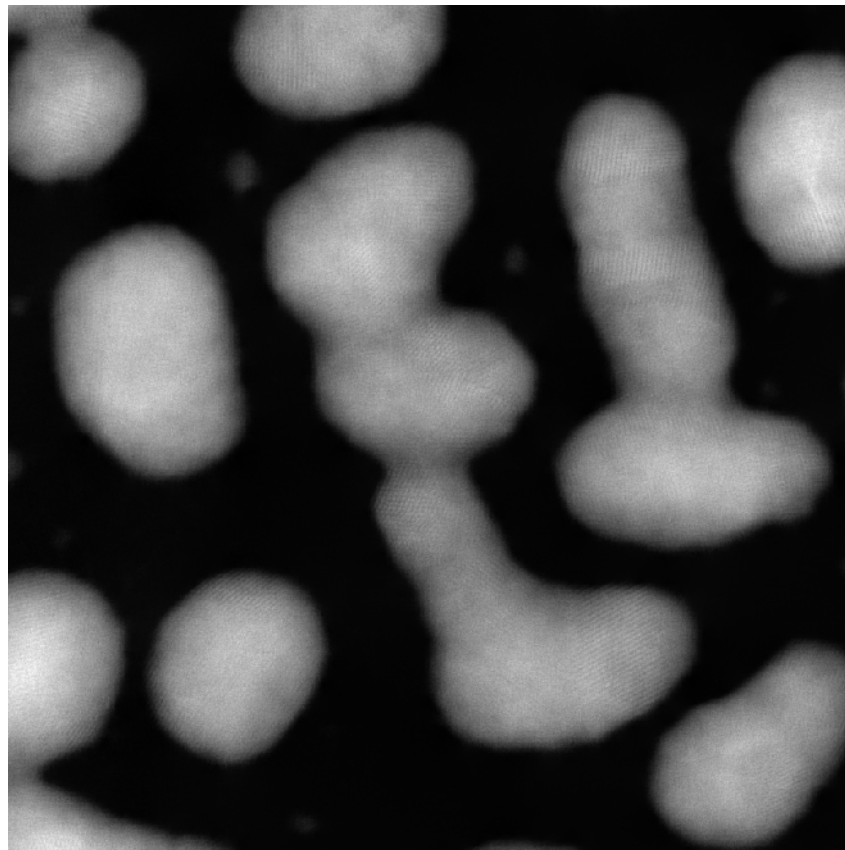


Adam $\sigma=2$ lr=auto iters=12
NCC 0.9772 MAE 0.1292 RMSE 0.2138 | nonrigid max shift 56.3 px | 0.1s

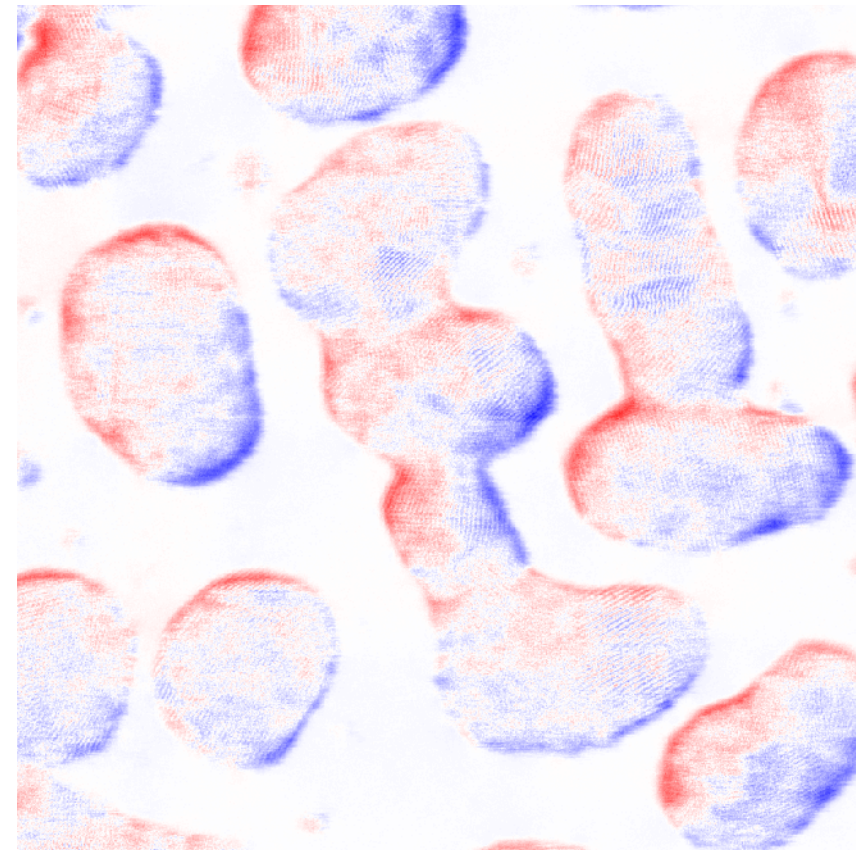
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

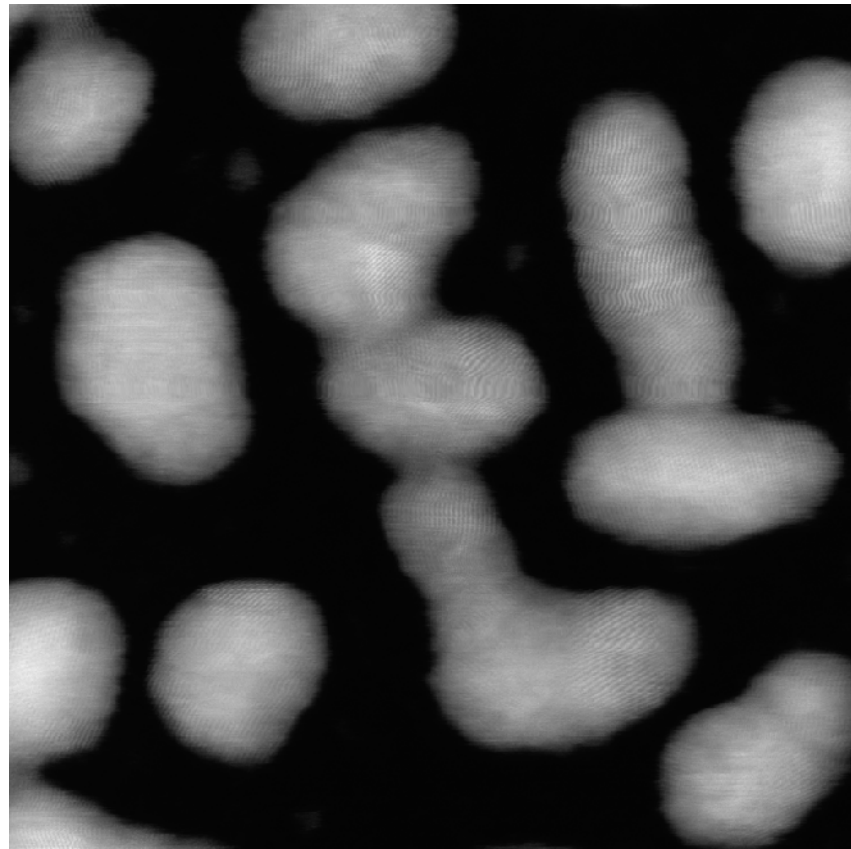


Difference: ref — corrected

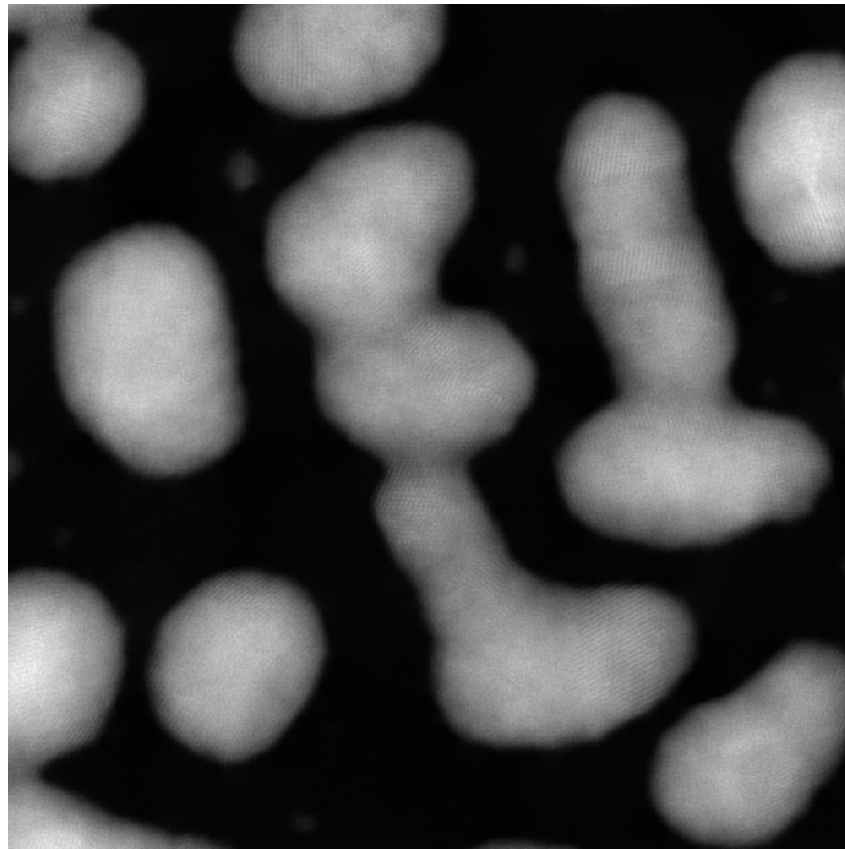


Adam $\sigma=2$ lr=0.1 iters=12
NCC 0.9816 MAE 0.1168 RMSE 0.1920 | nonrigid max shift 56.5 px | 0.1s

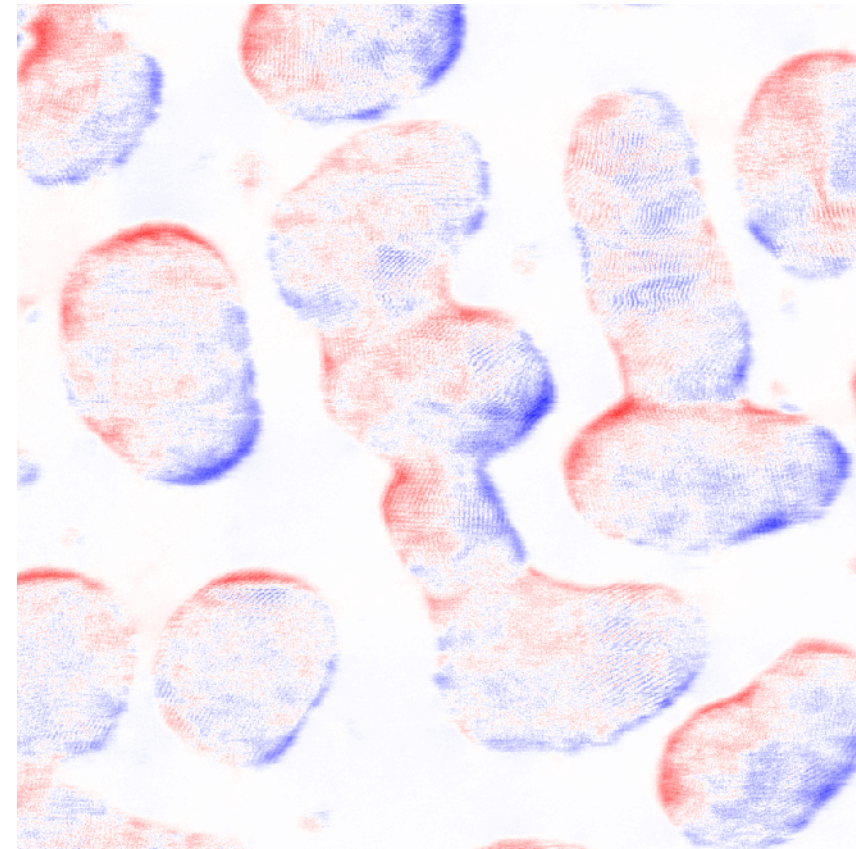
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

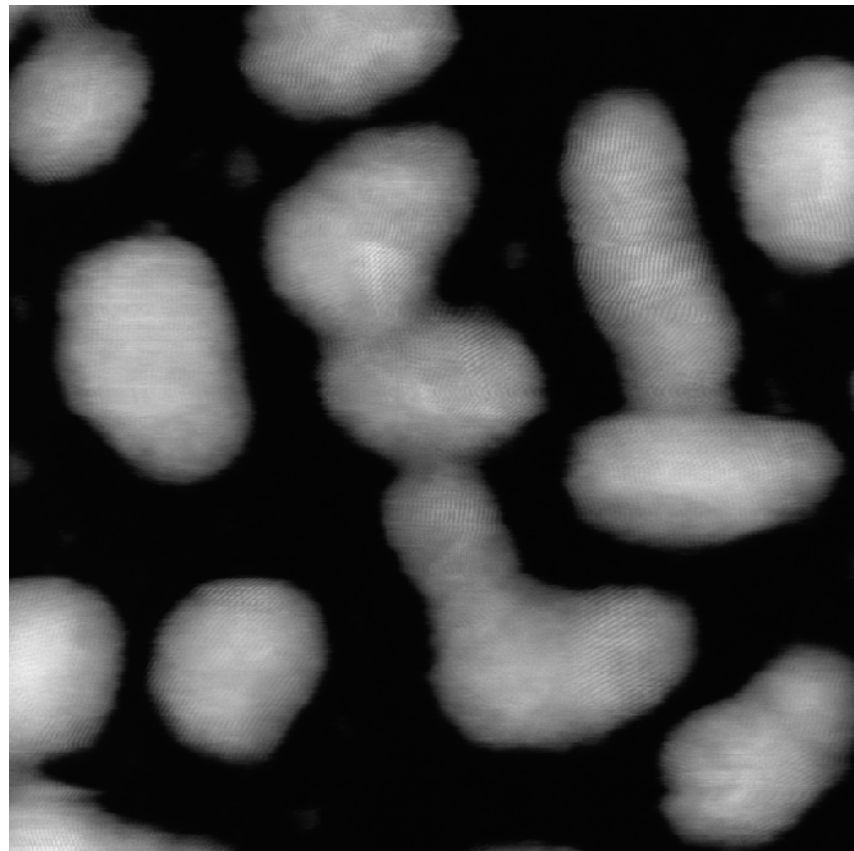


Difference: ref — corrected

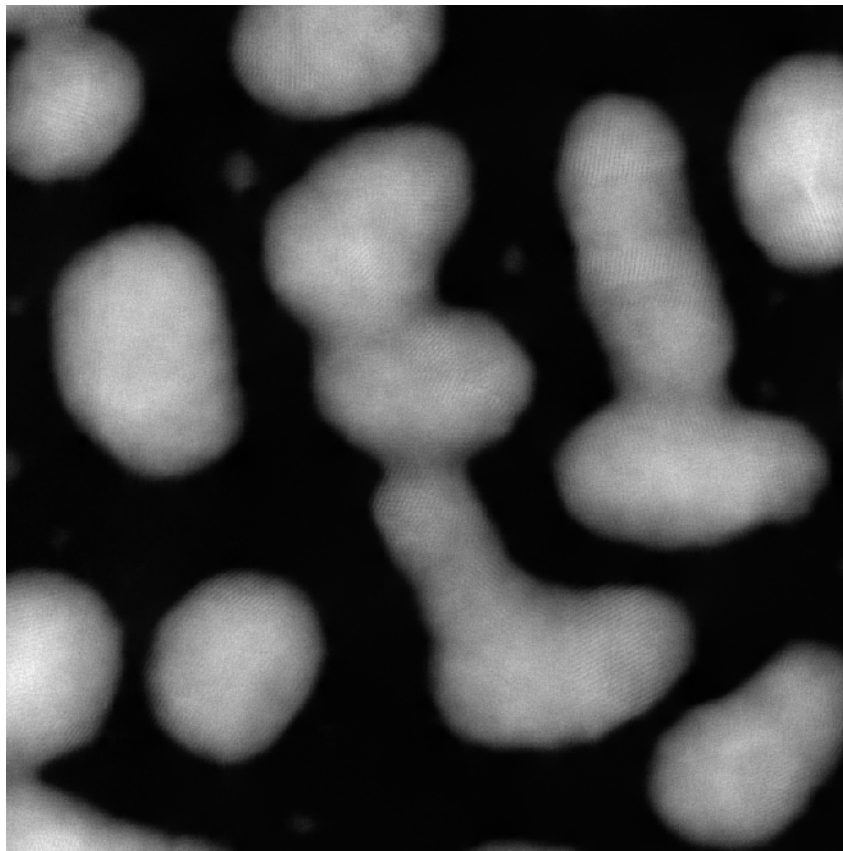


Adam $\sigma=2$ lr=0.5 iters=12 ← best
NCC 0.9847 MAE 0.1044 RMSE 0.1750 | nonrigid max shift 57.8 px | 0.1s

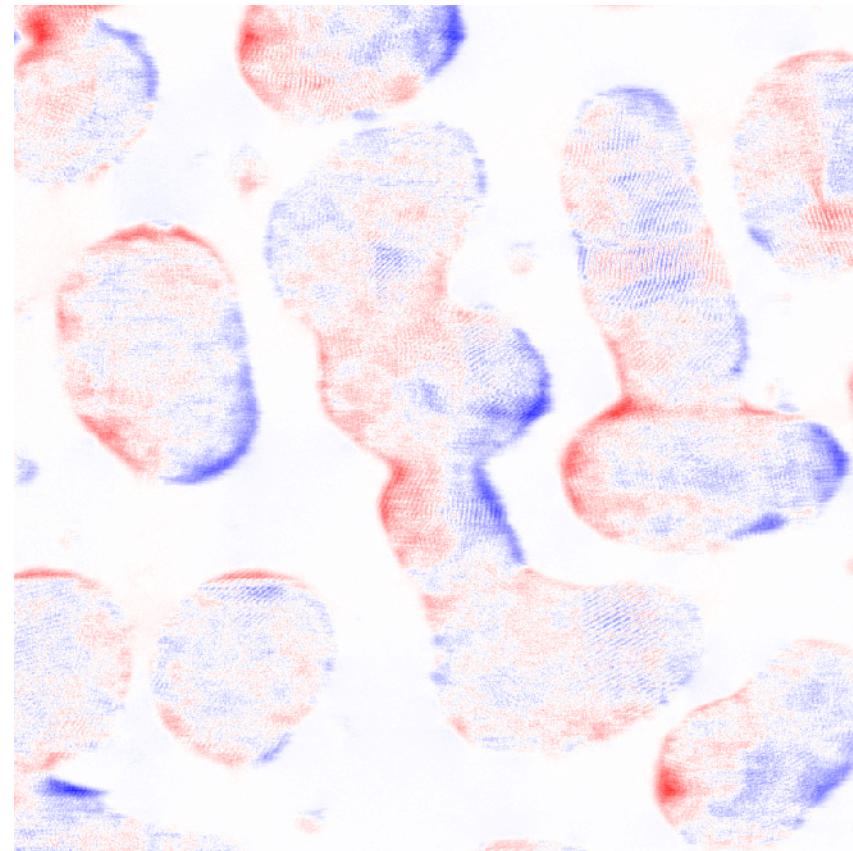
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

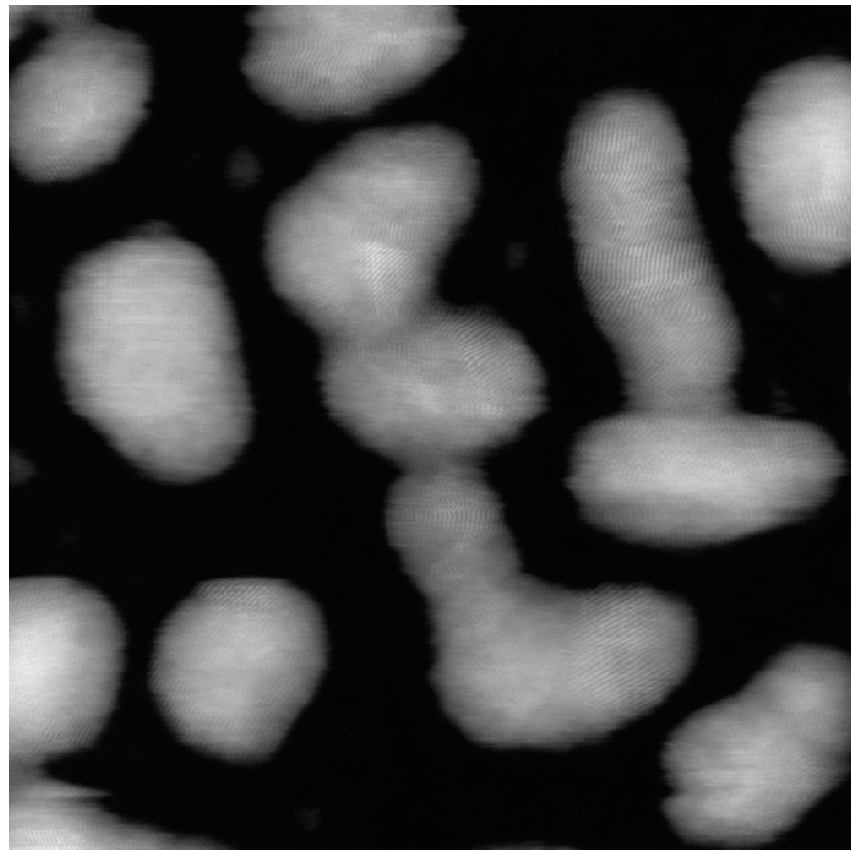


Difference: ref — corrected

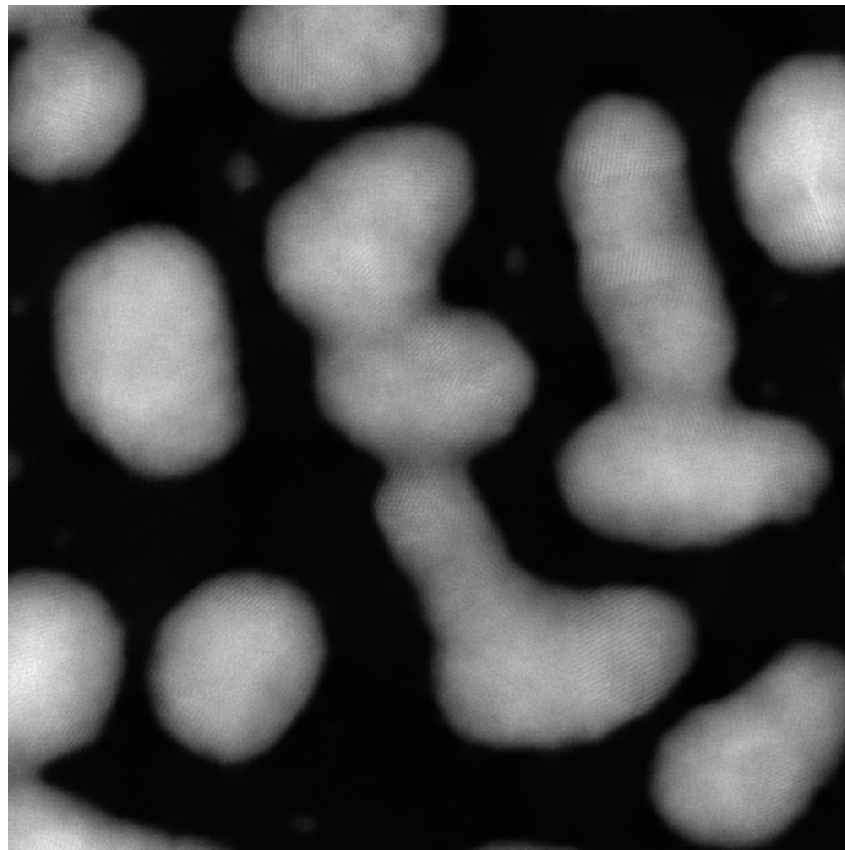


Adam $\sigma=2$ lr=1.0 iters=12
NCC 0.9806 MAE 0.1144 RMSE 0.1971 | nonrigid max shift 57.6 px | 0.1s

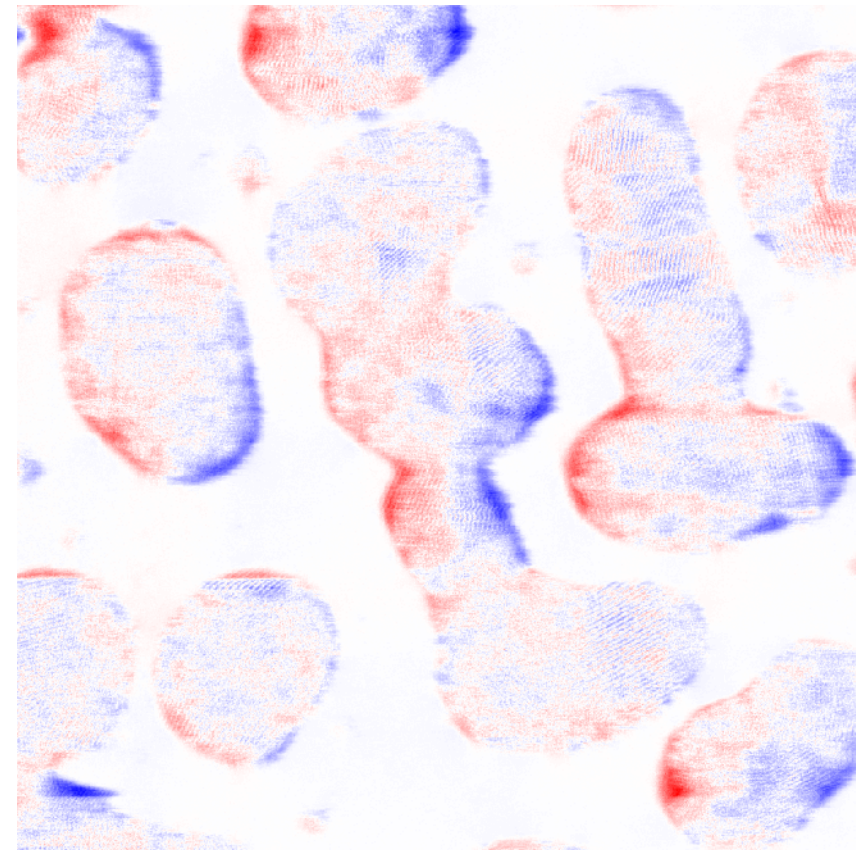
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)

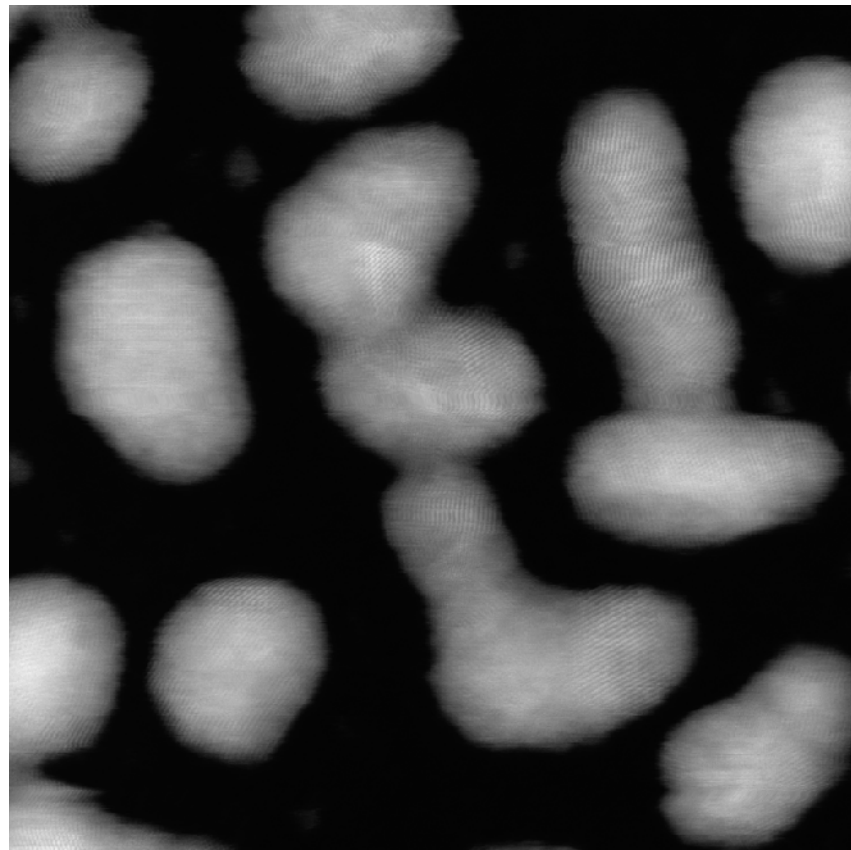


Difference: ref — corrected

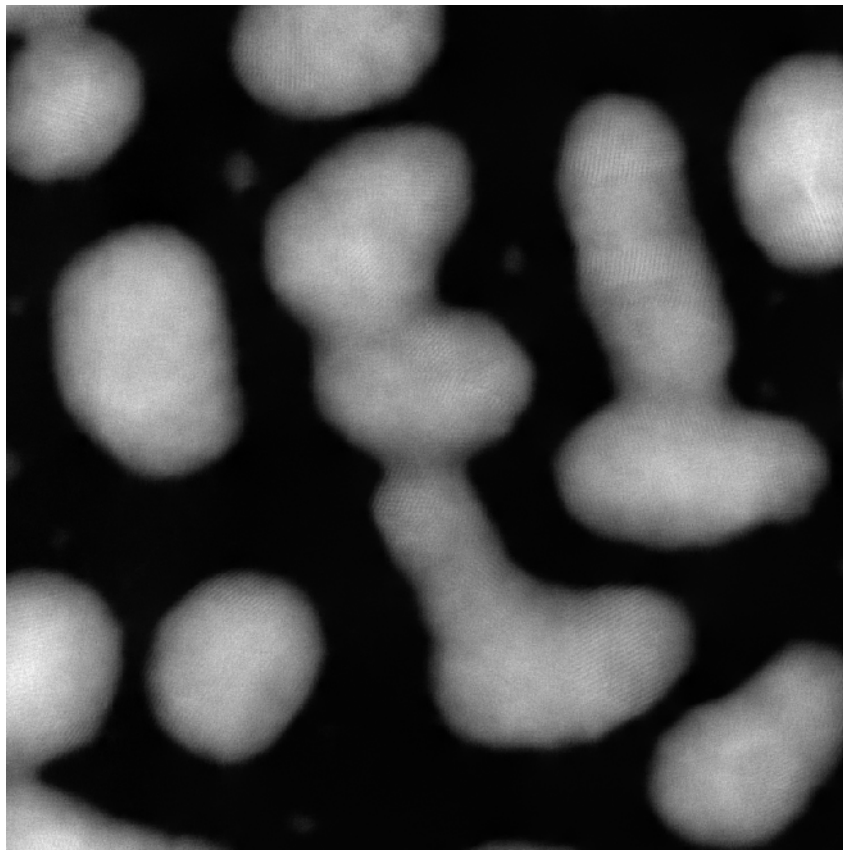


Adam $\sigma=2$ lr=0.5 iters=20 ★ BEST
NCC 0.9849 MAE 0.1039 RMSE 0.1740 | nonrigid max shift 57.2 px | 0.2s

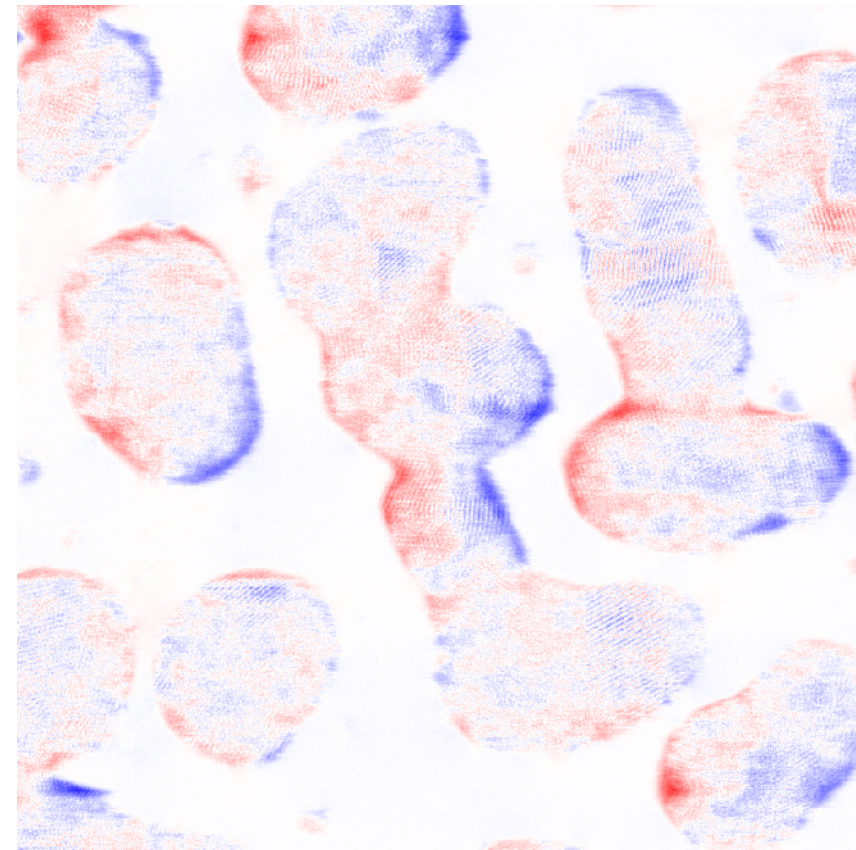
HAADF 0041 — drift-corrected



HAADF ref (0042+0043)



Difference: ref — corrected



Summary — All Trials
Reference: HAADF 0042+0043 (drift-corrected)

Config	NCC ↑	MAE ↓	RMSE ↓	Max shift	Time
raw	0.4080	0.7757	1.0881	—	—
affine	0.9718	0.1477	0.2373	—	—
$\sigma=8$ step=0.8 iters=12 (original)	0.9822	0.1122	0.1887	57.0 px	0.9s
LBFGS $\sigma=2$ step=0.8 iters=12	0.9840	0.1066	0.1791	58.6 px	0.4s
Adam $\sigma=2$ lr=auto iters=12	0.9772	0.1292	0.2138	56.3 px	0.1s
Adam $\sigma=2$ lr=0.1 iters=12	0.9816	0.1168	0.1920	56.5 px	0.1s
Adam $\sigma=2$ lr=0.5 iters=12	0.9847	0.1044	0.1750	57.8 px	0.1s
Adam $\sigma=2$ lr=1.0 iters=12	0.9806	0.1144	0.1971	57.6 px	0.1s
Adam $\sigma=2$ lr=0.5 iters=20	0.9849	0.1039	0.1740	57.2 px	0.2s